

2

D

Maximum velocity when $R = mg$

$$0.60 v = 3.0(9.81)$$

$$v = 49 \text{ m s}^{-1}$$

At 12 m s^{-1} , $F_{\text{net}} = mg - R = ma$

$$a = \frac{3.0(9.81) - 0.6(12)}{3.0} = 7.4 \text{ m s}^{-1}$$

3

C